Noor Islam S. Mohammad

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Research Interest

My research focuses on reliable and generalizable machine learning, with interests in deep learning, large language models, and vision-language models, particularly their reasoning, interpretability, and robustness.

Education

İstanbul Technical University (İTÜ), M.Sc. in Computer Science 2024–2027
Maslak, İstanbul, Türkiye • GPA: 3.20/4.00
Thesis: Grounding and Trust in Joint-Embedding Predictive World Models: Identifiability Limits and Calibrated Prediction Horizons
Research Advisor: Prof. Dr. Uluğ Bayazıt

University of the People, B.Sc. in Computer Science 2020–2023
Pasadena, CA, USA • GPA: 3.13/4.00

Cambridge International Education (British Council, Dhaka) 2018
GCE A-Levels: Further Mathematics (A*), Mathematics (A*), Physics (A*), Economics (A)

Publications and Manuscripts

⋆ denotes first and corresponding authorship, as reported for the listed works. Journal manuscripts identified as extended versions are cross-referenced to the related workshop papers. Publication status is stated separately for accepted workshop papers, manuscripts under review, and work in preparation. A complete list and available preprints are linked through Google Scholar.

Peer-Reviewed Workshop Papers (Accepted)

W1 Mohammad, N. I. S.⋆, & Bayazıt, U. (2026). SURGeLLM: Rethinking Multi-Task Evaluation through Task-Aware Feature Gating with Class-Balanced Normalization. *6th Workshop on Trustworthy NLP (TrustNLP), ACL 2026*, San Diego, CA, USA. [ACL Anthology]

W2 Mohammad, N. I. S.⋆ (2026). Trust-Aware Multi-Axis Reliability Auditing for NLP Classifiers: A Theoretical and Empirical Framework. *Workshop on Grounding Language Models: Learning Faithfully and Efficiently, EMNLP 2026*, Budapest, Hungary. [ACL Anthology]

W3 Mohammad, N. I. S.⋆, & Hasan, M. (2026). Coherence Under Commitment: Probing Generalization and Vacuous Memorization in LLM Logical Reasoning. *Workshop on Efficient Multimodal Question Answering (EMMQA), ICML 2026*, Seoul, South Korea. [OpenReview] [ICML]

W4 Mohammad, N. I. S.⋆, & Bayazıt, U. (2026). Salient Knowledge Pathways: Sparse Cross-Modal Routing for Efficient Knowledge-Intensive Multimodal Question Answering. *EMMQA, ICML 2026*, Seoul, South Korea. [OpenReview] [ICML]

W5 Mohammad, N. I. S.⋆, Hasan, M., Shammo, M. B. A. Z., Siddiki, H., & Habib, J. (2026). The Symmetry Budget: How Much Group Structure Should a Network Architecture Assume? *NeurIPS 2026 Workshop on Symmetry and Geometry in Neural Representations (NeurReps)*, Sydney, Australia. [NeurIPS], [OpenReview]

W6 Mohammad, N. I. S.⋆, Sheikh, M. F., & Habib, J. (2026). HALT: Certifying Trust Horizons in Action-Conditioned Robotic World Models. *NeurIPS 2026 Workshop on Robot Learning with World Models: Capabilities, Frontiers, and Challenges*, Sydney, Australia. [NeurIPS], [OpenReview]

W7 Mohammad, N. I. S.⋆, & Bayazıt, U. (2026). R^3DC: Reliability-Guided Reveal-to-Revise Depth Completion for Cross-Domain Sparse Perception. *CVPR 2026 Workshop on 3D Geometry Generation for Scientific Computing*, Denver, CO, USA. [OpenReview]

W8 Mohammad, N. I. S.⋆, & Bayazıt, U. (2026). CausalScene: Typed Causal Scene Graphs for Counterfactual Physical Reasoning with a Path to Video LLMs. *CVPR 2026 Workshop on Video Large Language Models*, Denver, CO, USA. [OpenReview]

W9 Mohammad, N. I. S.⋆, & Bayazıt, U. (2026). CausalScene: Learning Causal 3D Scene Graphs for Counterfactual Reasoning in Embodied Agents. *2nd Workshop on 3D Large Language Models and Vision-Language-Action Models (3D-LLM/VLA), CVPR 2026.* [OpenReview]

W10 Mohammad, N. I. S.⋆, & Bayazıt, U. (2026). Harmfulness Propagation Dynamics: Layer-wise Trajectories of Adversarial Intent in Large Language Models. *Mechanistic Interpretability Workshop, ICML 2026*, Seoul, South Korea. [OpenReview]

W11 Mohammad, N. I. S.⋆, & Sheikh, T. (2026). The Faithfulness Gap: Certifying Semantic Equivalence Between Natural-Language and Formal Mathematical Statements. *6th Muslims in Machine Learning (MusIML) Workshop, ICML 2026*, Seoul, South Korea. [OpenReview]

W12 Mohammad, N. I. S.*, & Sheikh, T. (2026). MIRAGE: Auditing Anti-Muslim Bias in Frontier LLMs Across Reasoning, Agentic, and Time-Coupled Conditions. *MusIML, ICML 2026*, Seoul, South Korea. [OpenReview]

Journal Manuscripts Under Review

J1 Mohammad, N. I. S.*, Bayazıt, U., & Hasan, M. (2026). CausalGraph: Typed Causal Scene Graphs for Inspectable Counterfactual Physical Reasoning. *International Journal of Computer Vision (IJCV)*. Substantially extended version of W8.

J2 Mohammad, N. I. S.*, & Bayazıt, U. (2026). HERALD: Harmfulness Propagation Dynamics—Layer-wise Trajectories of Adversarial Intent in Large Language Models. *Transactions on Machine Learning Research (TMLR)*. Substantially extended version of W10.

J3 Mohammad, N. I. S.*, & Bayazıt, U. (2026). R^3DC: Uncertainty-Aware Depth Completion via Reliability-Guided Cross-Modal Fusion and Adaptive Propagation. *Expert Systems with Applications*. Substantially extended version of of W7.

Conference Manuscripts Under Review

C1 Mohammad, N. I. S.*, Hasan, M., Sheikh, M. F., Hossain, M., & Sheikh, T. (2026). CausalScene: Explicit Intervention-Response Structure for Counterfactual Reasoning in Embodied Agents. *International Conference on Learning Representations (ICLR 2027)*. [OpenReview]

C2 Mohammad, N. I. S.*, Hasan, M., & Sheikh, M. F. (2026). NeuralGrad: Multi-Scale Gradient Reactivity for Open-World Network Intrusion Detection. *ICLR 2027*. [OpenReview]

C3 Mohammad, N. I. S.*, Habib, J., & Shammo, M. B. A. Z. (2026). Specialist Drift: Continual RL for Capability-Preserving Foundation Model Post-Training. *ICLR 2027*. [OpenReview]

C4 Mohammad, N. I. S.*, Shammo, M. B. A. Z., Siddiki, H., Hasan, M., Sheikh, M. F., & Habib, J. (2026). SIFT: Robust Meta-Faithfulness Verification of Chain-of-Thought Reasoning Under Distribution Shift. *ICLR 2027*. [arXiv] [OpenReview]

Manuscripts in Preparation

P1 Mohammad, N. I. S.*, Bayazıt, U., Habib, J., Sheikh, M. F., Sultana, S., & Moni, M. A. EthosVLA: Constitutional Decoding for Safety, Privacy, and Fairness in Frozen Embodied Agents. *[IJCV]*

P2 Mohammad, N. I. S.*, Sheikh, M. F., Bayazıt, U., & Sultana, S. HALT-JEPA: Certifying Trust Horizons in Action-Conditioned Robotic World Models. *[CVPR-2027 Main Track]*

P3 Mohammad, N. I. S.*, Sheikh, M. F., Siddiki, H., Habib, J., Uddin, R., Bayazıt, U., & Sultana, S. GAUGE-JEPA: Grounding is Gauge Fixing: Minimax Interaction Complexity for Controllable World Models Learned from Passive Video. *[CVPR-2027 Main Track]*

Research Experience

İTÜMultimedia Processing Lab — Maslak, İstanbul, Türkiye
Research Assistant (M.Sc. Thesis) — Nov 2025–Present

- Investigate representation identifiability and predictive reliability in joint-embedding predictive architectures, examining how latent-space structure affects the grounding and generalization of multimodal models.
- Develop action-conditioned predictive world-model frameworks that connect learned representations with anticipated state transitions, emphasizing physically grounded reasoning and temporally consistent prediction.
- Formulate uncertainty-calibration and reliability evaluations to characterize prediction-horizon stability, distribution-shift sensitivity, and failure conditions in vision-language and embodied prediction settings.

İTÜAI and Data Science Research Center — Maslak, İstanbul, Türkiye
Research Intern (Summer) — Jun 2025–Aug 2025

- Implemented end-to-end machine-learning workflows for heterogeneous datasets, covering data preprocessing, feature representation, model development, and experimental evaluation.
- Conducted comparative experiments across data regimes, analyzing predictive performance and model behavior through statistical evaluation and deep-learning-based approaches.
- Investigated methodological reliability through literature-driven experimental design, validation of evaluation procedures, and systematic documentation of empirical findings.

TÜBİTAK — İstanbul, Türkiye
Research Intern (Summer) — Jun 2024–Aug 2024

- Implemented and optimized PyTorch-based deep-learning models, translating experimental specifications into reproducible training and evaluation pipelines on established benchmarks.
- Performed controlled model evaluations and iterative experiments to investigate predictive behavior, compare architectural configurations, and assess model performance.

- Contributed to experimental validation and technical reporting, synthesizing implementation details, empirical observations, and research findings in interdisciplinary AI research activities.

Honors and Awards

- **Gold Reviewer Award**, International Conference on Machine Learning (ICML); top 25% of reviewers. 2026
- **STIC Incubator Program & Business Venture Seed Grant**, Startup Concept Competition; Round 1 top application, advanced to Round 2 video pitch. 2024
- **Türkiye Bursları Graduate Scholarship**, fully funded competitive international scholarship. 2023
- **TÜBİTAK Science Olympiad**, top 5% nationally among 8,000+ participants. 2023
- **IASC/NASA Citizen Science Asteroid Discovery Campaign**, Certificate of Recognition. 2021

Teaching and Research Mentoring

- **Mentor**, NeurIPS 2026 Affinity Group Mentorship Programme; provide research and submission guidance to an assigned participant.
- **Undergraduate Research Mentoring.** Mentored 10 undergraduate students in deep learning, computer vision, and NLP; four mentored projects resulted in workshop publications at ICML, NeurIPS, and CVPR.

Academic Reviewing and Service

- **Journals and publication venues:** Transactions on Machine Learning Research (TMLR, 2026); Springer Nature Multimedia Systems; IEEE Internet of Things (IoT); IEEE Computer Society.
- **Conferences:** ICML (2026); NeurIPS (2026); ACCV (2026); AAAI (2027); AISTATS (2027).
- **Workshops:** ICML (AI for Mathematics; Efficient Multimodal QA); ACL (Knowledge Foundation Models); CVPR (3D Geometry Generation; Foundation Models & Embodied Agents); NeurIPS (Symmetry & Geometry in Neural Representations; Robot Learning with World Models; VLMs for Real-World Deployment).

Technical Skills

- **Programming:** Python (advanced), C++, Java, SQL, Bash.
- **ML Systems:** PyTorch, TensorFlow, Hugging Face, CUDA, high-performance computing (HPC), multi-GPU training.
- **Methods:** Deep and representation learning, LLMs, VLMs, multimodal learning, reinforcement learning, generative models, retrieval-augmented generation, uncertainty quantification, calibration, conformal prediction, and hidden-state analysis.
- **Vision and Robotics:** OpenCV, sparse-depth perception, multi-sensor fusion, predictive world models, physically grounded reasoning.
- **Research Practice:** Experimental design, benchmarking, ablation studies, statistical evaluation, reproducibility, scientific writing.
- **Tools:** Linux, Git/GitHub, LaTeX, Overleaf, Jupyter/Colab, NumPy, SciPy, pandas, and Matplotlib.

Additional Academic Development

New York University, Tandon Bridge Programme 2024–2025
Computer Science Foundations, Online; passed with distinction.

American Council on Education (ACE) 2019–2024
58 ACE-endorsed college credits in Computer Science, verified through Coursera [Credly].

Selected Professional Certifications 2021–2025
Machine Learning • Deep Learning • Algorithms • Google IT Support • Software Developer • Writing in the Sciences.

Leadership and Languages

Microsoft Learn Student Ambassador (Gold) 2021–2024
Organized 15+ technical workshops reaching 500+ students, mentored students in technical learning, and participated in international student technology events, including Hack the North and HackMIT.

Languages: Bengali (native); English (fluent); Turkish (C1); Hindi and Urdu (professional working proficiency).

References

Prof. Dr. Uluğ Bayazıt, İstanbul Technical University, ulugbayazit@itu.edu.tr, [Faculty Profile]
Asst. Prof. Sharifa Sultana, University of Illinois Urbana-Champaign, sharifas@illinois.edu, [Website]
Prof. Mohammad Ali Moni, University of Queensland, m.moni@uq.edu.au, [Google Scholar]